

## Topic to discuss

→ Newton divided difference Interpolation.

→ Numerical Problem -2

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## Some Key Concepts

If  $(x_0, y_0), (x_1, y_1), (x_2, y_2)$  be given points, the first divided difference for  $x_0, x_1$  is defined by the relation,

$$[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0} = \Delta y_0$$

Similarly,  $[x_1, x_2] = \frac{y_2 - y_1}{x_2 - x_1} = \Delta y_1$

Second divided difference of  $x_0, x_1, x_2$  is defined

$$\text{as } [x_0, x_1, x_2] = \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_0} = \frac{\Delta y_1 - \Delta y_0}{x_2 - x_0} = \Delta^2 y_0$$

Similarly, third divided difference of  $x_0, x_1, x_2, x_3$

$$\text{is defined as } [x_0, x_1, x_2, x_3] = \frac{[x_1, x_2, x_3] - [x_0, x_1, x_2]}{x_3 - x_0}$$

for  $n^{\text{th}}$  divided difference,

$$y[x_0, x_1, \dots, x_n] = \frac{y[x_1, x_2, \dots, x_n] - y[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0}$$

## Method of Induction

Method of mathematical induction is a proof technique used to show that a statement is true for all natural numbers (usually  $n = 1, 2, 3, \dots$ )

It works in two main steps:

### 1. Base case

Prove that the statement is true for the first value (usually  $n = 1$ )

### 2. Induction step

Assume that statement is true for some arbitrary natural number  $n = k$  (this is called induction hypothesis) and then prove that it is true for  $n = k + 1$

If both the statement is true, then the statement is true for all natural numbers.

## Principle of Mathematical Induction

Let  $P(n)$  be a statement defined for all natural numbers  $n \in \mathbb{N}$   
If

- 1)  $P(1)$  is true (base case) and
- 2) for some  $k \in \mathbb{N}$ , the truth of  $P(k)$  implies the truth of  $P(k+1)$  (Induction step),

then  $P(n)$  is true for all natural numbers  $n \geq 1$ .

## Numerical Problem : 2

Q:- Given the function  $y = \frac{1}{x}$ , show that the divided difference of  $n^{\text{th}}$  order

$$y[x_0, x_1, x_2, \dots, x_n] = (-1)^n / (x_0 x_1 x_2 \dots x_n)$$

Solution: By the definition of divided difference

for any function  $y = f(x)$

$$\Delta f(x) = \Delta y = f[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

similarly, for  $n^{\text{th}}$  order,

$$\Delta^n f(x) = \Delta^n y = f[x_0, x_1, x_2, \dots, x_n] = \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, x_2, \dots, x_{n-1}]}{x_n - x_0}$$

We have,  $y = f(x) = \frac{1}{x}$

so,  $y_0 = f(x_0) = \frac{1}{x_0}$

Applying method of Induction to prove this,

for  $n=1$ ,

$$\Delta f(x) = \Delta y = y[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0} = \frac{\frac{1}{x_1} - \frac{1}{x_0}}{x_1 - x_0}$$

$$= \frac{-1}{x_0 x_1}$$

Similarly,

$$\Delta f(x) = y[x_1, x_2] = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\frac{1}{x_2} - \frac{1}{x_1}}{x_2 - x_1}$$

$$= \frac{-1}{x_1 x_2}$$

for  $n=2$ ,

$$\begin{aligned}\Delta^2 f(x) &= \Delta^2 y = y[x_0, x_1, x_2] = \frac{y[x_1, x_2] - y[x_0, x_1]}{x_2 - x_0} \\&= \frac{\frac{-1}{x_1 x_2} - \left(-\frac{1}{x_0 x_1}\right)}{x_2 - x_0} \\&= \frac{-1}{x_2 - x_0} \left( \frac{x_0 - x_2}{x_0 x_1 x_2} \right) \\&= \frac{(-1)^2}{x_0 x_1 x_2}\end{aligned}$$

Assume the statement is true for order  $n-1$ .

That means :

$$y[x_0, x_1, \dots, x_{n-1}] = \frac{(-1)^{n-1}}{x_0 x_1 \dots x_{n-1}}$$

$$y[x_1, x_2, \dots, x_n] = \frac{(-1)^{n-1}}{x_1 x_2 \dots x_n}$$

Use the recursive formula for order  $n$

$$\begin{aligned}y[x_0, x_1, \dots, x_n] &= \frac{y[x_1, x_2, \dots, x_n] - y[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0} \\&= \frac{1}{x_n - x_0} \left[ \frac{(-1)^{n-1}}{x_1 x_2 \dots x_n} - \frac{(-1)^{n-1}}{x_0 x_1 x_2 \dots x_{n-1}} \right] \\&= \frac{(-1)^{n-1}}{x_n - x_0} \left[ \frac{x_0 - x_n}{x_0 x_1 x_2 \dots x_n} \right] \\&= \frac{(-1)^{n-1} x - 1}{x_0 x_1 x_2 \dots x_n} = \frac{(-1)^n}{x_0 x_1 x_2 \dots x_n}\end{aligned}$$

Hence,

$$y[x_0, x_1, \dots, x_n] = \frac{(-1)^n}{x_0 x_1 \dots x_n}$$

(Proved)  
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